

UNIT 1 • MACHINE LEARNING FOUNDATIONS

Data and Datasets

Understanding the raw material of every machine learning system

T.Y.B.C.A. | Semester V



What we will cover

1

What is data?

Definition, properties, real examples

2

Types of data

Structure, representation, labelling

3

Data to knowledge

The DIK progression

4

Splitting data

Train / validation / test sets

5

Pros & challenges

Why data helps, where it hurts

6

What is a dataset?

Rows, columns, features, target

7

Types & properties

Numerical, image, time series...

8

Operations in Python

pandas & NumPy workflow

9

Data vs Dataset vs DB

Clear comparison

10

Applications

Where all of this is used

CONCEPT

What is Data?

Data is a set of observations or measurements — raw facts, values, text, sounds or images that have not yet been interpreted or analysed.

Why it matters

- ML algorithms simply cannot be trained without data
- Model performance depends on both quality and quantity of data
- AI, automation and analytics are all powered by large data collections

Real-world example

\$19 billion

Facebook's purchase of WhatsApp — driven largely by access to user data that could improve its services.

“Algorithms are the engine — data is the fuel.”

Properties of Data



Volume

Scale generated every millisecond



Variety

Text, images, video, audio, health



Velocity

Speed of generation & streaming



Value

Meaningful insight it can yield



Veracity

Accuracy and reliability



Viability

Fitness for use in a system



Security

Protection from tampering



Accessibility

Easy to reach for decisions



Integrity

Consistent through its lifecycle



Usability

Simple to read and interpret

The first three are the classic "3 V's" of Big Data; the rest describe data quality and governance.

Types of Data — based on Structure

Structured

Stored in rows and columns —
spreadsheets and relational databases.

EXAMPLES

Sales records, customer details, financial
transactions

TYPICAL USE

Supervised learning: regression &
classification

Semi-structured

Has organisational markers but does not
fit neatly into a table.

EXAMPLES

JSON, XML, NoSQL documents

TYPICAL USE

Web scraping, API responses, social
media analysis

Unstructured

No predefined format, so it is the
hardest to process.

EXAMPLES

Text files, images, video, audio

TYPICAL USE

NLP, image recognition, speech-to-text

Types of Data — Representation & Labelling

Based on Representation

1

Numerical

Measured in numbers

`age = 21, income = 45000`

2

Categorical

Names, labels or groups

`gender, fruit type, city`

3

Ordinal

Categories with a natural order

`Small < Medium < Large`

Based on Labelling

1

Labelled data

Inputs together with the correct target output.

`age + income → loan approved / rejected`

Used by supervised learning

2

Unlabelled data

Only input variables, no target attached.

`a folder of photos with no captions`

Used by unsupervised learning

From Data to Knowledge

DATA

Raw, unprocessed facts, values, text, sound or images. Not yet interpreted.



INFORMATION

Data that has been processed, organised and interpreted so it carries meaning.



KNOWLEDGE

Information combined with experience and insight — the basis for good decisions.

Worked example — a retail store

- The store collects written customer feedback forms — this is raw data.
- Analysing the forms shows recurring themes such as pricing and product quality — this is information.
- Acting on those themes to redesign the product range builds knowledge the business can reuse.

Where ML Data Comes From



Healthcare

Patient records, laboratory results, medical imaging



Finance

Transaction logs, credit history, fraud reports



E-commerce

Product reviews, purchase history, browsing clicks



Transportation

GPS traces, traffic reports, sensor readings



Social Media

Posts and images, engagement metrics, network graphs

Every one of these domains feeds supervised, unsupervised or recommendation models.

HOW FAST IS DATA GROWING?

The numbers behind the hype

40 ZB

Data generated by 2020 —
a 300x rise since 2005

161 bn GB

Produced by the medical
sector in 2011 alone

400 mn

Tweets sent each day by
200 million users

27%

Of organisational data is
inaccurate, hurting trust

More than 4 billion hours of video are watched every month, and users post about 30 billion pieces of content in the same period.

How Data is Split in Machine Learning

1 Training data

The model learns patterns from this portion. It sees the answers here.

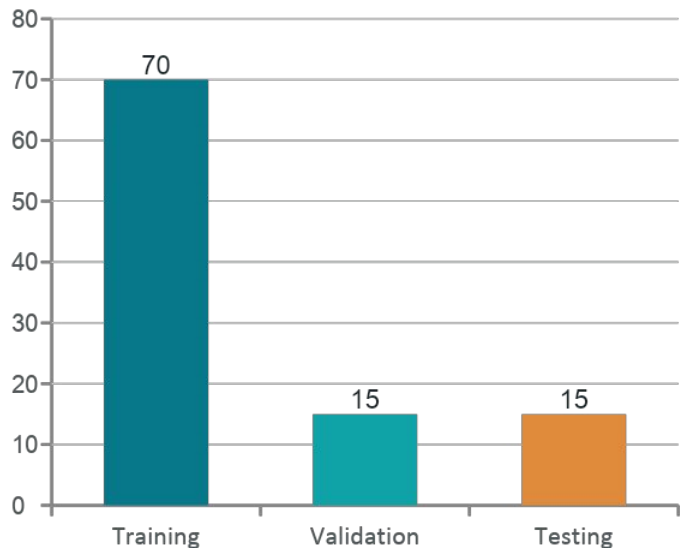
2 Validation data

Used during training to tune hyper-parameters and decide when to stop.

3 Testing data

Held back until the end to measure how well the model generalises.

A typical 70 / 15 / 15 split



Splitting Data in Python

```
from sklearn.model_selection import train_test_split
import pandas as pd
```

```
df = pd.read_csv("students.csv")
```

```
X = df.drop("result", axis=1)    # features
y = df["result"]                # target
```

```
# first split: 80% train+val, 20% test
X_temp, X_test, y_temp, y_test = train_test_split(
    X, y, test_size=0.20, random_state=42)
```

```
# second split: carve validation out of temp
X_train, X_val, y_train, y_val = train_test_split(
    X_temp, y_temp, test_size=0.1875,
    random_state=42)
```

```
print(len(X_train), len(X_val), len(X_test))
```

Points to remember

- `random_state` fixes the shuffle so results are reproducible.
- Never evaluate on data the model has already trained on.
- Use `stratify=y` when classes are imbalanced.
- For small datasets prefer k-fold cross validation.

Advantages of Using Data in ML

1 Improved accuracy

With more data, algorithms can capture more complex relationships between inputs and outputs, improving prediction and classification.

2 Automation

Models handle repetitive work and decision-making faster and more consistently than people can.

3 Personalisation

User data lets systems tailor recommendations and experiences to each individual.

4 Cost savings

Automating routine tasks reduces manual effort and raises overall efficiency.

WATCH OUT FOR THESE

Challenges in Using Data for ML



Data quality

Noisy, incomplete or irrelevant records lead to weak models.



Data quantity

Too few examples and the model cannot generalise.



Bias & fairness

A skewed dataset reproduces and amplifies discrimination.



Overfitting

The model memorises training data and fails on new inputs.



Underfitting

The model is too simple to capture the underlying pattern.



Privacy

Misusing sensitive data creates legal and ethical problems.



PART 2

Datasets



Structure, types, properties and how we work with them in Python

DEFINITION

What is a Dataset?

A **dataset** is a structured collection of related data, usually organised as rows and columns, describing one particular category or domain.

A student dataset, for instance, keeps one row per student and one column per attribute such as name, age, grade and marks.

Remember

Rows = records / observations. Columns = features / attributes. One column is usually the target we want to predict.

Example: students.csv

Roll	Name	Age	Marks	Result
101	Aditi	20	78	Pass
102	Rahul	21	42	Fail
103	Sneha	20	91	Pass
104	Imran	22	65	Pass

4 records × 5 features — “Result” is the target variable.

Why Datasets Matter

1 Analysis

They supply the raw material needed for analysis and decision making.

2 Training models

They make it possible to train and test ML and AI models.

3 Discovering patterns

They reveal patterns, correlations and insights across domains.

4 Innovation

They support R&D in healthcare, finance, education and more.

5 Benchmarking

Shared datasets allow reproducibility and fair comparison of results.

Types of Datasets

Numerical

Numeric points analysed with statistics

daily temperature readings

Categorical

Discrete groups or classes

colour, gender, occupation

Time series

Values recorded over a period

daily stock prices

Ordered

Ranked values where order matters

survey ratings, movie rankings

Image

Pictures for recognition tasks

X-rays for disease detection

Web

Pulled from APIs or web sources

JSON responses, scraped pages

File based

Stored in ordinary files

CSV, Excel, plain text

Note

One project often mixes several types — an e-commerce model uses numerical, categorical, image and time-series data together.

Statistical Properties of a Dataset

Centre

The “middle” of the data, measured by mean, median or mode.

```
df.mean(), df.median()
```

Skewness

How symmetrical the distribution is. A normal curve has skewness 0.

```
df.skew()
```

Spread

How far points sit from the centre — variance and standard deviation.

```
df.std(), df.var()
```

Outliers

Points far outside the general pattern that can distort results.

```
IQR or z-score test
```

Correlation

How strongly two variables move together, positively or negatively.

```
df.corr()
```

Distribution

Normal, uniform, binomial — decides which tests are valid.

```
sns.histplot(df.x)
```

Features of a Dataset



Numerical features

Quantities such as height or weight; may be continuous or discrete.



Categorical features

Values belonging to classes such as gender or colour.



Size of the data

The number of entries (rows) and features (columns) the file holds.



Data entries

The individual values stored inside the dataset.



Target variable

The one feature we want to predict or explain using the others.

```
>>> df.shape
```

```
(4, 5)
```

```
>>> df.columns
```

```
Index(['Roll', 'Name',  
      'Age', 'Marks', 'Result'])
```

```
>>> df.dtypes
```

```
Roll      int64  
Name      object  
Age       int64  
Marks     int64  
Result    object
```

```
>>> df['Result'] # target
```

Operations on Datasets in Python

1. Loading & reading

Bring data in from CSV, JSON, SQL or an API.

`read_csv()` `read_json()` `read_excel()`

3. Preprocessing

Clean, fill missing values, encode and scale.

`drop()` `fillna()` `dropna()` `copy()`

5. Visualisation

Plots and charts that expose hidden patterns.

`plot()` `hist()` `scatter()`

7. Exporting

Save the processed result for later use.

`to_csv()` `to_json()` `to_excel()`

2. Exploratory analysis

Summaries, distributions and first impressions.

`head()` `tail()` `info()` `groupby()`

4. Manipulation

Filter, group, merge and reshape the data.

`merge()` `concat()` `join()`

6. Indexing

Access rows and columns by label or position.

`loc[]` `iloc[]` `set_index()`

Loading and Exploring a Dataset

```
import pandas as pd

df = pd.read_csv("students.csv")    # 1. load

print(df.head())                   # 2. explore
print(df.info())
print(df.describe())

df = df.drop_duplicates()           # 3. clean
df["Marks"] = df["Marks"].fillna(df["Marks"].mean())

pass_df = df[df["Result"] == "Pass"] # 4. filter
print(df.groupby("Result")["Marks"].mean())

df.to_csv("students_clean.csv", index=False)
```

What each step gives you

- `head()` — a quick look at the shape
- `info()` — dtypes and null counts
- `describe()` — centre and spread

Try it yourself

- Iris and Titanic from seaborn
- `load_digits()` from scikit-learn
- Open data portals such as data.gov.in

Data vs Dataset vs Database

Aspect	Data	Dataset	Database
Definition	Raw facts with no context	Structured collection of related entries	Organised store of many datasets
Structure	Unorganised	Rows and columns	Tables, often linked together
Role	Backbone of everything else	Gives structure and insight	Defines relationships between entities
Handling	Cannot be used directly	Analysed with Python, Power BI, Tableau	Queried with SQL, transactions, scripts
Usage	Needs preprocessing first	Analysis, modelling, visualisation	Querying and running applications

In short: data → the ingredient, dataset → the prepared dish, database → the whole kitchen store.

Challenges in Working with Datasets

Quality issues

Inconsistent or carelessly collected records mislead the analysis and lower model accuracy.

validate at
collection time

Missing data

Gaps in records make conclusions unreliable and confuse training.

impute or drop
carefully

Bias

Unbalanced or unrepresentative samples produce unfair outcomes.

check class
distribution

Scalability

Very large datasets are hard to clean, store and process on limited hardware.

chunking, Dask,
sampling

Privacy

Personal data must be handled under strict security and legal rules.

anonymise,
restrict access

Applications of Datasets

1 Machine learning training

Image recognition, natural language processing and fraud detection all begin with a labelled dataset.

2 Business analytics

Sales, customer and financial datasets drive pricing, stocking and strategy decisions.

3 Healthcare

Patient records, scans and genomic data support diagnosis and treatment planning.

4 Education & research

Public datasets let students and researchers test hypotheses and build new solutions.

5 Recommendation systems

Clicks, purchases and ratings are used to suggest products, films and music.

KEY TAKEAWAYS

What you should remember

- Data is raw observation; once processed it becomes information, and applied information becomes knowledge.
- Data is classified by structure (structured, semi-structured, unstructured), by representation (numerical, categorical, ordinal) and by labelling.
- A dataset is data organised into rows (records) and columns (features), usually with one target variable.
- Datasets are split into training, validation and testing sets so that performance is measured on unseen data.
- pandas and NumPy cover the whole workflow: load, explore, preprocess, manipulate, visualise, index and export.
- Quality, bias, missing values, scale and privacy are the recurring challenges — and better data usually beats a better algorithm.

References & Practice

Practice questions

- Define data. Explain any six properties of data.
- Differentiate structured, semi-structured and unstructured data with examples.
- Explain the progression from data to information to knowledge with an example.
- Why is a dataset split into training, validation and testing sets?
- Compare data, dataset and database in tabular form.
- List and explain any five operations performed on datasets using pandas.

Assignment

Download any public CSV dataset, load it with pandas, report its shape, dtypes and missing values, and plot one feature.